

When does the plan score matter? OGBench Cube

A companion to the TwoRoom study: the same plan scores on a world model whose latent is *already adequate*

OGBench cube-single · LeWM-cube, DINO-WM-cube · goal-conditioned CEM-MPC

Setup

We repeat the TwoRoom plan-score comparison on OGBench **cube-single** (a 7-DoF arm must move a cube to a target). We use the benchmark’s own protocol: goal-conditioned MPC evaluated by replaying expert episodes and setting the goal to the state k steps ahead (*goal-offset* k), success = cube within the goal radius. Two frozen world models: **LeWM-cube** (pretrained) and **DINO-WM-cube** (a predictor we trained; see caveat). Plan scores: *latent* ($\|\hat{z} - z_g\|^2$, the default), *TRM* (regression metric), and *TD value* (offline goal-conditioned temporal-difference, quasimetric head). All numbers are % of 40 goals reached; reference points: **expert** $\approx 99\%$, published baselines **LeWM 74** / **DINO 86**, perfect planner 100. (Sanity: on 8 short goals every method scores 100%.)

1. Plan-score matrix

World model	goals	latent	TRM	TD value
LeWM-cube	easy (offset 25)	80.0	82.5	85.0
LeWM-cube	hard (offset 50)	70.0	72.5	72.5
DINO-cube	easy (offset 25)	82.5	80.0	82.5
DINO-cube	hard (offset 50)	72.5	75.0	72.5

All four conditions sit in a tight band; the largest spread (LeWM easy: latent 80 $\rightarrow$ TD 85, i.e. +2 successes in 40) is within $n=40$ sampling noise ($\pm \sim 7\%$). Learned metrics neither clearly help nor hurt.

2. TD-settings sweep (offset 25)

	expectile τ (at $n=20$)				n -step (at $\tau=0.1$)		
	0.1	0.3	0.5	0.7	1	5	20
LeWM	85	85	80	80	85	85	85
DINO	82.5	85	87.5	87.5	82.5	85	82.5

TD hyperparameters barely move success (all 80–87.5%). The expectile — the knob that *doubled* hard-goal success on TwoRoom — is inert here; LeWM mildly prefers low τ , DINO mildly high, both within noise.

3. Planning-budget sweep (LeWM): does a weaker planner expose the plan score?

CEM budget	easy (offset 25)		hard (offset 50)	
	latent	TD value	latent	TD value
10×3 (≈ 30 evals)	80.0	85.0	72.5	72.5
30×5 (≈ 150)	85.0	85.0	70.0	70.0
60×8 (≈ 480)	82.5	82.5	72.5	70.0
150×15 (full)	80.0	85.0	70.0	72.5

No. Even a *tiny* 10×3 planner already reaches $\sim 80\text{--}85\%$, and the TD-minus-latent gap stays $\sim 0\text{--}5$ (noise) at every budget. On TwoRoom a weak planner exposed the broken latent; on cube the latent is good enough that minimal search suffices, so the plan score never becomes the bottleneck.

4. Offline vs. online TD (LeWM, offset 25)

	offline	online $\tau 0.1$	online $\tau 0.3$	online $\tau 0.5$	online $\tau 0.7$
success %	85.0	80.0	82.5	82.5	82.5

Online TD (the offline value continued on $\sim 2,500$ planner-collected latents) $\approx$ offline at every expectile — no gain. On TwoRoom online TD helped because the offline value was poor on the planner’s (broken-latent, hard cross-wall) distribution; here the offline value is already good, so self-practice adds nothing.

5. Twocube (two-cube manipulation)

Does a *harder* task push the latent back into the regime where the plan score pays off? We repeat the comparison on OGBench **cube-double** (both cubes must reach targets) with DINO-WM-double (a predictor we trained — there is no pretrained double-cube world model, and no LeWM-double).

A methodological note first. Our initial run rolled the predictor 25 steps open-loop and every plan score returned an *identical* 37.5% — a red flag. The cause: this simplified residual-MLP predictor **diverges** beyond ~ 5 steps (terminal $\|z\|$: $1.1 \rightarrow 5 \times 10^5$ at 10 steps $\rightarrow$ non-finite at 25), so the costs were garbage and equal only by coincidence. Following DINO-WM’s short-horizon *receding* MPC, we retrain and roll only **5 steps** (the predictor’s stable range), which gives the valid numbers below.

	latent	TRM	TD value	online TD
DINO-WM-double ($n=80$)	31.2	30.0	31.2	32.5

The scores are now **statistically indistinguishable** ($30\text{--}32.5\%$; spread $\ll$ the $\pm \sim 5.5\%$ band at $n=80$, though the success *sets* differ, so the plan score does change behaviour). Success caps at $\sim 31\%$ — far below single-cube’s $\sim 80\%$ — because the short-horizon, simplified double-cube WM is **capability-limited**: it solves only $\sim 31\%$ of two-cube goals regardless of terminal cost, leaving no room for the plan score to help. Twocube is thus **inconclusive** on the plan-score question; settling it needs a stronger double-cube model (the paper’s $H=3$ ViT predictor, or a trained LeWM-double).

Bottom line

Across five axes — learned metric vs. latent, TD hyperparameters, goal difficulty, planning budget, and offline-vs-online — single-cube success never leaves the $\sim 70\text{--}87\%$ band. This is the clean **inverse of TwoRoom**, where a broken latent (16%) made the learned reachability metric decisive ($\rightarrow 68\%$). On the harder **twocube** the scores again cluster ($\sim 31\%$), but for a different reason — a capability-limited world model, not an adequate latent. The unifying principle: **the learned plan score is decisive in exactly one regime — a *misleading* latent with an *otherwise-capable* model+planner (TwoRoom). When the latent is already good (cube-single) or the model itself is the bottleneck (twocube), the plan score is inert.**

Details. Eval = World.evaluate goal-offset replay, num_eval 40, success via env. Plan-score matrix / sweeps at CEM 150×15 unless noted. TD value = offline goal-conditioned temporal-difference, quasimetric head, $n=20$, $\gamma=0.99$. **Caveat:** DINO-WM-cube is a quick $\sim 6k$ -step predictor we trained (action_block 1), not a tuned checkpoint — its row is the weaker leg; LeWM-cube is the pretrained model and is the more reliable line. Twocube: DINO-WM-double rolled 5 steps (receding), $n=80$; the predictor is stable only for ≤ 5 -step rollouts (the paper uses an $H=3$ ViT predictor). $n=40/80 \Rightarrow$ noise $\pm \sim 7\%/5.5\%$.